

Figure 1 — Cytokine Comparison with Model-Based p-values

Data Science for Biology Workshop

Data Import & Merge

Load Model Results

Figure 1 — Cytokine Comparison with BH-Adjusted p-values

```
# =====  
# Boxplots of log-transformed cytokine concentrations by PCOS status  
# Each panel annotated with Benjamini-Hochberg adjusted p-value  
# from the linear mixed-effects model  
# =====  
  
# Compute y-position for each cytokine panel (place label at top-right)  
anno_positions <- analysis_data %>%  
  group_by(cytokine) %>%  
  summarise(  
    y_max = max(log_conc, na.rm = TRUE),  
    y_range = diff(range(log_conc, na.rm = TRUE)),  
    .groups = "drop"  
  ) %>%  
  left_join(pval_anno, by = "cytokine") %>%  
  mutate(  
    x = 1.5,  
    y = y_max + y_range * 0.10  
  )  
  
# Create the boxplot  
p <- analysis_data %>%  
  ggboxplot(x = "pcos_status", y = "log_conc",
```

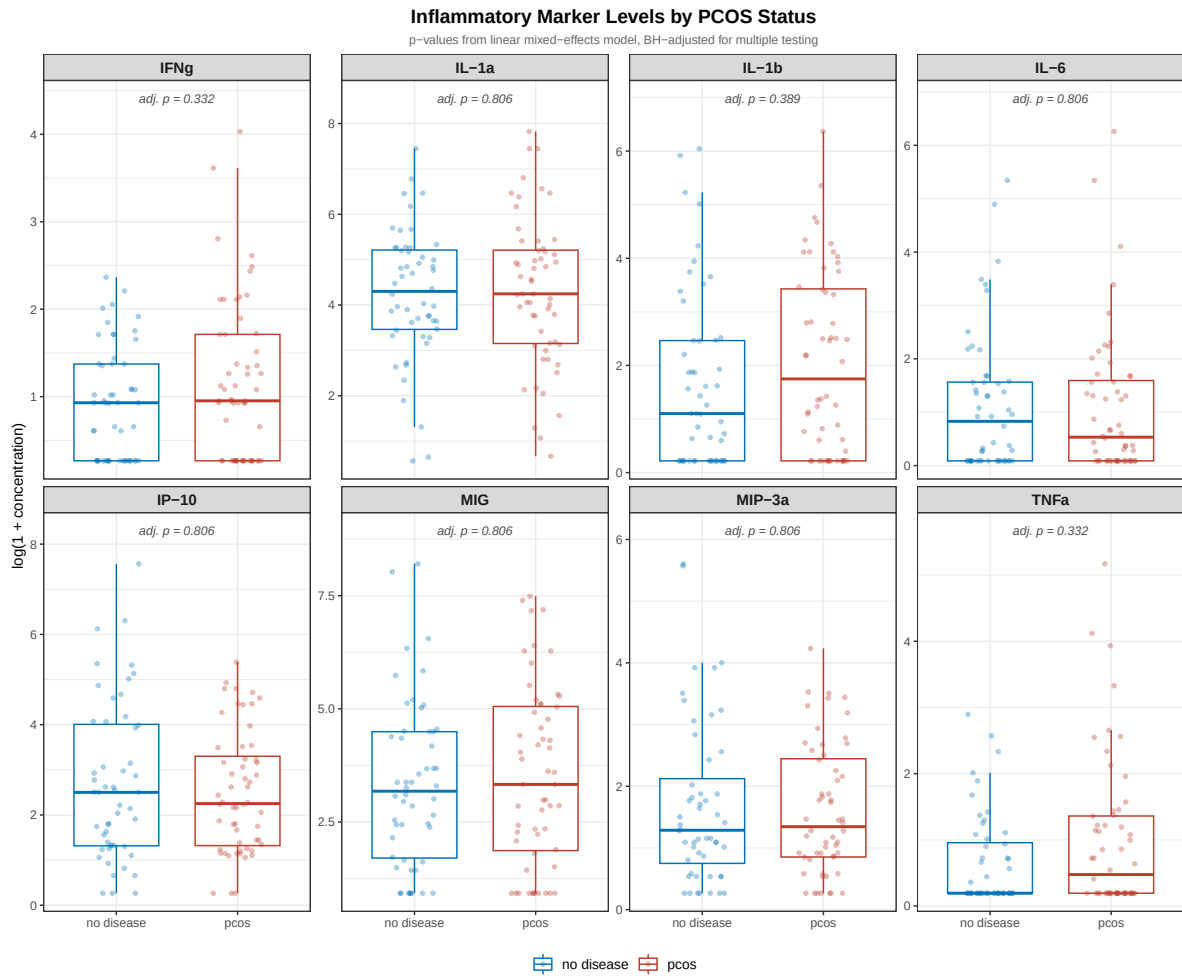
```

        color = "pcos_status",
        palette = c("#0072B5", "#BC3C29"),
        add = "jitter",
        add.params = list(alpha = 0.35, size = 1.2)) +
facet_wrap(~ cytokine, scales = "free_y", ncol = 4) +
labs(title = "Inflammatory Marker Levels by PCOS Status",
      subtitle = "p-values from linear mixed-effects model, BH-adjusted for multiple testing",
      x = NULL,
      y = "log(1 + concentration)") +
theme_bw() +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold", size = 14),
  plot.subtitle = element_text(hjust = 0.5, size = 9, colour = "grey40"),
  strip.text = element_text(face = "bold", size = 11),
  axis.title.y = element_text(size = 11),
  legend.title = element_blank(),
  legend.text = element_text(size = 10),
  legend.position = "bottom"
)

# Add one clean p-value annotation per panel
p <- p + geom_text(
  data = anno_positions,
  aes(x = x, y = y, label = label),
  size = 3.2,
  colour = "grey30",
  fontface = "italic",
  inherit.aes = FALSE
)

print(p)

```



```
# Save figure
ggsave("Figure_1_Cytokine_Comparison_with_pvalues.pdf",
       plot = p, width = 12, height = 10, dpi = 300)
```